higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK

NQF LEVEL 4

(12030504)

22 November 2019 (X-Paper)
(09:00-12:00)

This question paper consists of 5 pages.

Copyright reserved

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

Copyright reserved

QUESTION 1

1.1 Give a short explanation of the following terms when constructing support structures for geysers. Write only the answer next to the question number (1.1.1 – 1.1.5) in your answer book.

1.1.1 Support deck

1.1.2 Tie-beam

1.1.3 Drip tray

1.1.4 Joist

1.1.5 Webs

(5 × 1) (5)

1.2 A geyser is heavy and can cause serious damage to a roof construction and the rest of the home if not correctly installed and supported. The additional member used to strengthen the tie beam must be fitted strictly in accordance with the requirements stipulated.

Make use of the key words below and discuss the installation requirements of a geyser support deck, for a geyser with a capacity less than 250 litres.

nailing method; grade; extend; recommended length; competent person; minimum trusses required

(8)

1.3 Certain types of roof designs can make it difficult for workmen to operate in because of the limited space.

State TWO reasons why it is not permissible to remove or cut away the webs of a roof truss when installing a support deck for a geyser. (2)

[15]

QUESTION 2

2.1 Justify the benefits and reasons for ceilings in buildings. (12)

2.2 What is the standard permissible distance between brandering for the support framework of a gypsum ceiling and give ONE reason? (2)

2.3 What is the purpose of packing material when fixing the battens for ceilings? (2)

2.4 State THREE benefits of installing cornice moulding where the end of the ceiling meets the walls of the room. (3)

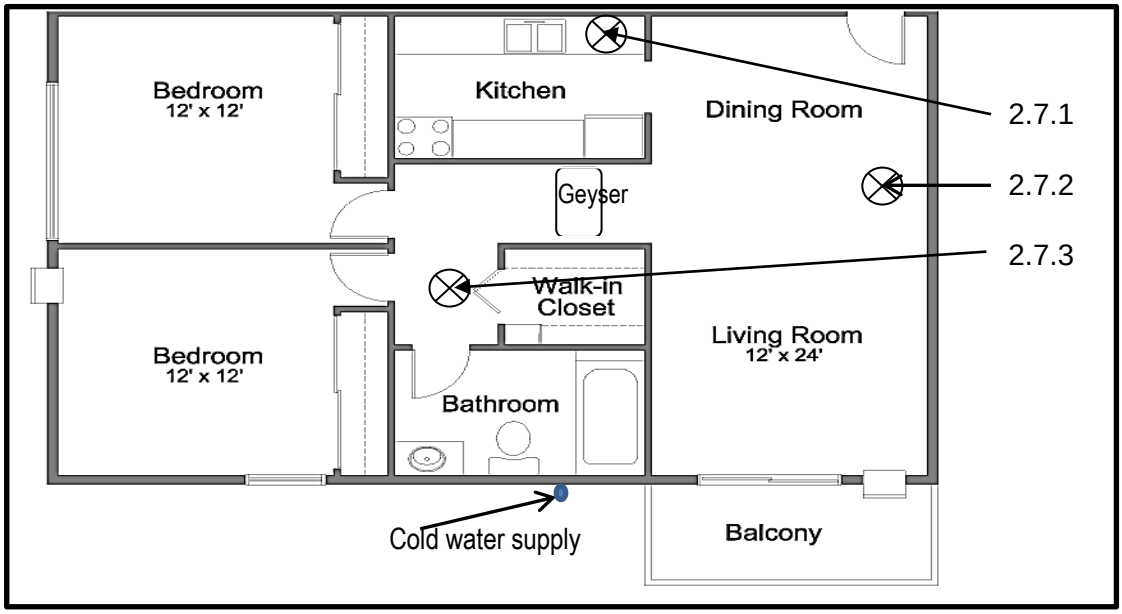
2.5 What is the minimum size of a trap door? (1)

2.6 What is the minimum weight a trap door must support? (1)

Copyright reserved

2.7 FIGURE 1 below shows the plan view of a two bedroom house with a hipped roof structure. The plan is numbered on the right hand side (2.7.1-2.7.3) that indicate possible positions for a trap door.

Give TWO reasons why you will or why you will not install a trap door at each of the positions indicated as (2.7.1-2.7.3) in FIGURE 1. Write only the answer next to the question number (2.7.1-2.7.3) in the ANSWER BOOK.



Source: *www.biteinto.info*

FIGURE 1

(3 × 2) (6)

2.8 Name THREE types of material commonly used to insulate a ceiling. (3)

[30]

QUESTION 3

3.1 Compare the advantages and disadvantages between suspended timber ground floors and concrete floors. (8)

3.2 State THREE reasons why laminated timber beams are preferable to natural timber beams when used in constructing suspended timber floors. (3)

3.3 Explain why good ventilation (airflow) beneath a suspended timber floor is essential. (3)

3.4 Give brief descriptions of the following terms used in the construction of suspended timber floors:

3.4.1 Sleeper wall (2)

3.4.2 Ant guard (2)

3.4.3 Damp proof membrane (2)

[20]

QUESTION 4

4.1 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write 'true' or 'false' next to the question number (4.1.1–4.1.5) in the ANSWER BOOK. Correct the statement if it is FALSE.

4.1.1 Outside doors are usually heavier than inside doors because they are made from solid timber.

4.1.2 A doorframe should be in a shape of a parallelogram with all corners 90°.

4.1.3 If the jambs of a doorframe are slightly misaligned, you could tweak it by moving the hinge on the contact side at least 6 mm closer to the rebate.

4.1.4 If the floor is not level and runs out of level downwards, away from the door, you must notify the supervisor and request that the bricklayer fix the problem before you can fit and hang the door.

4.1.5 Doors mostly installed at the front of houses or offices are fitted with glass panels to allow light in.

(8)

4.2 State the SIX steps in the correct order a carpenter takes when hanging a new door. (6 × ½) (3)

4.3 Discuss the procedures that you will follow to fit a two lever mortise lock in a new door. (9)

[20]

QUESTION 5

5.1 State THREE advantages of using manufactured boards rather than solid timber when building kitchen cabinets. (3)

5.2 Explain the purpose of a plinth of a cupboard and the reason why it should be made of solid pine when installed in a kitchen. (4)

5.3 A profile gauge is used during the installation of pre-manufactured cupboards.

Explain how the profile gauge works and how it can be used to accurately mark cut-outs in kickboards that must intersect with skirting boards and quadrants. (4)

5.4 Why is it advisable to fit the handles only after the pre-manufactured cupboards have been delivered to the site for installation? (2)

5.5 State TWO disadvantages of open shelving. (2)

[15]

TOTAL: 100

Copyright reserved